

Optimal Correction of Environmental Stochastic Externalities

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Abstract

Pollution raises climate risk in two ways. It can raise the probability of a discrete catastrophe, the "tipping point" already central to the climate-economics literature. It also shifts the entire distribution of everyday climate shocks, making extreme weather more frequent and more intense. We study this distributional channel and its consequences for the social cost of carbon (SCC). We introduce stochastic externalities: random damages whose distribution has parameters that depend on the pollution stock rather than staying fixed. Solving the social planner's problem, first in a three-period model and then in an infinite-horizon recursive one, we show that the optimal carbon tax adds a correction term to the standard Pigouvian tax, equal to expected marginal damages weighted by the distribution's sensitivity to pollution. The correction admits closed forms for standard distribution families and nests the Pigouvian tax as a special case. Under heavy-tailed shocks it can grow without bound faster than the Pigouvian term, sharpening Weitzman's Dismal Theorem. A calibrated DICE exercise raises the SCC by [X%].

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JEL Classification: *D62; D82; H21; H23*

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1 Introduction

Human activity raises climate risk in two ways: it makes a climate tipping point more likely, and it shifts the distribution of everyday weather events, making them more intense and less predictable. We focus on this second mechanism, known in the Earth sciences but largely unexplored in economics. Its costs are already large and growing. In the United States, the annual number of billion-dollar weather and climate disasters averaged 8.5 over 1980–2023 but 20.4 over the most recent five years, 2019–2023 (NOAA National Centers for Environmental Information (NCEI), 2024). Using NOAA simulation data, Bakkensen and Barrage (2026) finds that climate change will roughly triple welfare losses from cyclones in the United States and the Caribbean.

In this paper, we develop a framework to study how a distributional shift affects the social cost of carbon (SCC). We model what we call stochastic externalities: random climate shocks drawn from a distribution whose parameters depend on the pollution stock rather than being fixed values. To build intuition, we first enhance a three-period neoclassical growth model in which current emissions harm a fraction of future output. The resulting SCC is the sum of a standard Pigouvian term and a stochastic externality correction. The correction can be written in several equivalent ways. The simplest is the quantile form: expected future marginal damages, weighted by how much the pollution-induced shift moves each quantile of the shock distribution. Put differently, marginal damages are weighted by how much more or less likely each shock realization becomes after the shift.

We then work through three examples that translate this correction into economically meaningful channels. Under normally distributed shocks, the correction splits into three channels: a mean-shift channel, a risk-aversion channel, and a damage-curvature channel. Under Pareto shocks, it collapses to a single tail-fattening channel, an expectation of future damages weighted by a term that grows faster than linearly. This term can grow without bound, and faster than the standard Pigouvian term, reinforcing Weitzman’s concerns about a Dismal Theorem. The log-normal case lies between the two. It inherits the normal’s three channels (each now weighted by the shock level) and adds a Jensen channel. Although it never strictly diverges, it can still dominate the Pigouvian term. The framework is not limited to these cases: it admits any parametric family of continuous distributions satisfying mild regularity conditions, and it nests the standard SCC when shocks are exogenous. In a calibrated numerical example, we show the stochastic externality raises the SCC by [X%] above the Pigouvian level.

Afterwards, we generalize the framework to a recursive infinite-horizon model. The standard recursive framework invokes Markov properties to justify a recursive form (see for example Stokey et al., 1989); here we appeal instead to Markov *Decision* Process arguments, which let the past state affect not only the current state but also the transition distribution. We then apply the framework to a modified DICE benchmark, showing how the stochastic externality channel moves the SCC in practice. This calibration is deliberately conservative: it uses a thin-tailed, symmetric normal distribution, which understates the effect that thicker-tailed shocks would produce.

This paper is not the first to model endogenous climate risk. A large economics literature treats climate change as raising the probability of a discrete catastrophe, typically by modeling

its arrival as a Poisson process whose rate increases with pollution (Barro, 2015; Brausmann & Bretschger, 2024; Cai & Lontzek, 2019; Müller-Fürstenberger & Schumacher, 2015; van der Ploeg, 2014; van der Ploeg & de Zeeuw, 2018; Zemel, 2015). That literature addresses the graver risk: a possibly civilizational or existential regime shift. We address a more mundane channel, in which the entire shock distribution shifts gradually as pollution accumulates. Unlike a tipping point, this channel is already costly and already underway, but it has drawn far less attention from economists. It is a matter not only of uncertain timing but of uncertain impacts. We model it by letting the whole distribution of climate shocks move continuously with the pollution stock, capturing the steady degradation of the climate risk profile rather than the odds of a sudden regime shift. A parallel empirical literature is beginning to estimate how climate change reshapes the distribution of weather shocks and their damages (Bakkensen & Barrage, 2026), so this paper connects this micro evidence to the macro question of optimal policy.

Section 2 introduces the three-period model, moving from standard externalities to stochastic externalities and closing with the normal, Pareto, and log-normal examples. Section 3 generalizes the analysis to multiple periods, and Section ?? applies it to a modified DICE benchmark. Section ?? concludes.

2 Three-period model

2.1 Standard externalities

Consider the simplest case, in which there are only three periods and output Y contributes to a pollution stock S that reduces a fraction $D(S) \in [0, 1)$ of next period's total output. Suppose damages are increasing and convex in the pollution stock. The planner solves:

$$\begin{aligned} \max_{\{c_0, c_1, c_2, k_1, k_2, S_1, S_2\}} \quad & u(c_0) + \beta u(c_1) + \beta^2 u(c_2) \\ \text{s.t.} \quad & \end{aligned}$$

$$\begin{aligned} c_0 + k_1 - (1 - \delta)k_0 &= (1 - D(S_0))Y(k_0) \\ c_1 + k_2 - (1 - \delta)k_1 &= (1 - D(S_1))Y(k_1) \\ c_2 - (1 - \delta)k_2 &= (1 - D(S_2))Y(k_2) \\ S_1 &= (1 - \phi)S_0 + Y(k_0) \\ S_2 &= (1 - \phi)S_1 + Y(k_1) \end{aligned}$$

This problem has the following modified Euler condition between periods 0 and 1.

$$u'(c_0) = \beta u'(c_1) \left(1 - \delta + \left[1 - D(S_1) - \underbrace{\beta \frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2)}_{\text{"Standard" externality}} \right] Y'(k_1) \right)$$

Since this is a standard problem, we do not solve the household's problem in detail. The household's Euler condition does not contain the "standard" externality above, so the planner would like to set a carbon tax $\tau = \beta \frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2)$ on output to internalize it. This is the standard Pigouvian tax, equal to the discounted marginal damages of pollution. Since second-period output is entirely determined by first-period investment, only the Euler condition between periods 0 and 1 is affected.

We also consider the case in which damages depend on the pollution stock as well as a random shock x following an "exogenous" distribution, $D(S, x)$. Suppose damages are increasing and convex in both the pollution stock and climate shocks. The Euler condition then becomes the following, implying an optimal carbon tax $\tau = \beta \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2) \right]$.

$$u'(c_0) = \beta \mathbb{E}_{x_1} \left[u'(c_1) \left(1 - \delta + \left[1 - D(S_1, x_1) - \underbrace{\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2) \right]}_{\text{"Standard" externality}} \right] Y'(k_1) \right) \right]$$

2.2 Stochastic externalities

Now consider the same model, but with future damages stochastic and the parameters of their distribution shifting with the pollution stock. We make the following assumptions about the stochastic process: (i) the shocks follow a parametric statistical model that (ii) has fixed support, (iii) has a pdf and cdf continuous and differentiable in the parameter θ , (iv) has a parameter $\theta(S)$ continuous and differentiable in the pollution stock S , and (v) has a strictly increasing cdf. The social planner solves:

$$\begin{aligned} \max_{\{c_0, c_1, c_2, k_1, k_2, S_1, S_2\}} \quad & u(c_0) + \beta \int u(c_1(x_1)) f(x_1; \theta(S_1)) dx_1 \\ & + \beta^2 \int \int u(c_2(x_2)) f(x_1; \theta(S_1)) f(x_2; \theta(S_2)) dx_1 dx_2 \\ \text{s.t.} \quad & \end{aligned} \tag{1}$$

$$\begin{aligned} c_0 + k_1 - (1 - \delta)k_0 &= (1 - D(S_0))Y(k_0) \\ c_1(x_1) + k_2(x_1) - (1 - \delta)k_1 &= (1 - D(S_1, x_1))Y(k_1) \quad \forall x_1 \\ c_2(x_2) - (1 - \delta)k_2(x_1) &= (1 - D(S_2, x_2))Y(k_2(x_1)) \quad \forall x_1, x_2 \\ S_1 &= (1 - \phi)S_0 + Y(k_0) \\ S_2 &= (1 - \phi)S_1 + Y(k_1) \end{aligned}$$

We write the expectations in integral form to make explicit the dependence of the distribution on the pollution stock, which enters through the PDF $f(x; \theta(S))$. Meanwhile, the consumer solves the following problem, denoting by θ_1 and θ_2 the parameters of the distributions for x_1 and x_2 respectively.

$$\begin{aligned} \max_{\{c_0, c_1, c_2, k_1, k_2\}} \quad & u(c_0) + \beta \int u(c_1(x_1)) f(x_1; \theta_1) dx_1 + \beta^2 \int \int u(c_2(x_2)) f(x_1; \theta_1) f(x_2; \theta_2) dx_1 dx_2 \\ \text{s.t.} \quad & \end{aligned}$$

$$\begin{aligned} c_0 + k_1 - (1 - \delta)k_0 &= (1 + r_0)k_0 \\ c_1(x_1) + k_2(x_1) - (1 - \delta)k_1 &= r_1(x_1)k_1 \quad \forall x_1 \\ c_2(x_2) - (1 - \delta)k_2(x_1) &= r_2(x_2)k_2(x_1) \quad \forall x_1, x_2 \end{aligned}$$

The consumer takes the PDF as exogenous and so does not treat θ as a function of pollution. This leads to the following modified expressions for the optimal corrective tax.

Proposition 2.2.1 (Three-period carbon tax with endogenous stochastic externalities).
The planner can decentralize the solution to problem (1) with the following carbon tax:

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} - \underbrace{\beta \text{Cov} \left(\frac{u(c_2)}{u'(c_1)}, \frac{\partial \ln(f(x_2; \theta(S_2)))}{\partial S_2} \right)}_{\text{"Stochastic" externality}}$$

This tax can alternatively be expressed as follows:

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} + \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_x(S_2, x_2) Y(k_2) \frac{\partial Q(p(x_2); \theta(S_2))}{\partial S_2} \right]}_{\text{"Stochastic" externality}}$$

Proof. See Appendix A for full proof. □

The setup nests the standard externality. When shocks are degenerate, the tax reduces to the deterministic Pigouvian tax; when shocks are exogenous, it reduces to the Pigouvian tax in expectation. With stochastic externalities, equation (5) carries an additional term that raises the corrective tax when it is negative.

The first equation in Proposition 2.2.1 has a positive second term when utility and the score move in opposite directions on average. Utility is monotone decreasing in shocks, since higher shocks mean lower output and therefore lower consumption. The score, meanwhile, can be positive or negative depending on whether a given shock realization becomes more or less likely after the shift.

The second equation is equivalent but reads differently. The first factor is the stochastic discount factor, the second is the marginal damage of shocks on output, and the third is the change in the quantile function caused by the shift. For example, take the third quartile: the term says whether it moves up, so the threshold below which three-quarters of shocks fall is now higher, or down.

To build intuition on how the extra term works, we present three illustrative examples. The framework applies to any distribution satisfying the assumptions stated at the start of the section.

Normal shocks. Let $x \sim N(\mu(S), \sigma(S))$ such that $\mu'(S) > 0$ (higher pollution increases average shocks) and $\sigma'(S) > 0$ (higher pollution increases shock variability).

Proposition 2.2.2 (Normal distribution decomposition).

Under normal shocks, the stochastic externality correction term decomposes into three channels:

$$\begin{aligned} -\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= \mu'(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right] \\ &\quad - \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right)^2 \right] \\ &\quad + \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 D(\cdot)}{\partial x_2^2} Y(k_2) \right] \end{aligned}$$

Proof. See Appendix B for full proof. □

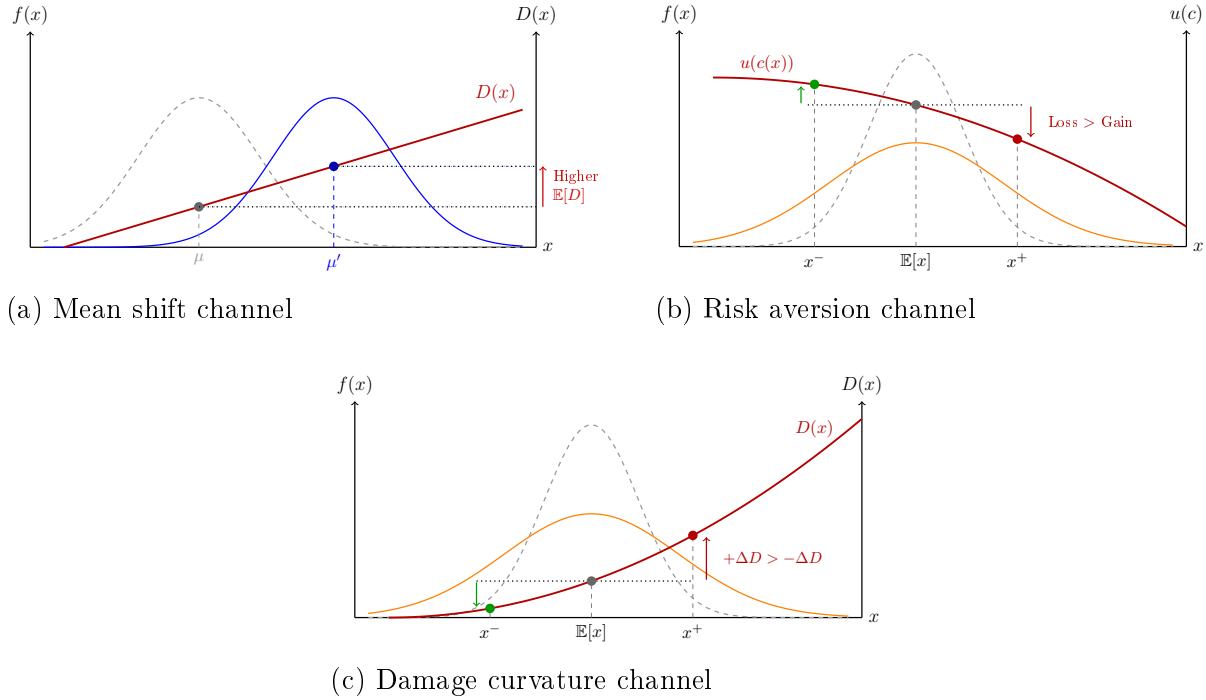


Figure 1: Stochastic externality channels for normally distributed shocks.

Figure 1 gives intuition for the three channels. The mean-shift channel in panel (a) raises the carbon tax when damages are increasing, since climate shocks are now higher on average. The risk-aversion channel in panel (b) raises the tax when utility is concave: a variance increase makes both low and high shocks more likely, but the utility loss in the right tail exceeds the gain in the left tail. If utility is linear, such that the agent is risk-neutral, this

channel vanishes because the agent weighs the extra gain and the extra loss equally. The damage-curvature channel in panel (c) depends on the curvature of the damage function with respect to climate shocks. If damages are convex, such that they grow faster at larger shocks, this channel raises the tax because a wider distribution adds more damage in the right tail than it removes in the left.

Pareto shocks. Let $x \sim \text{Pareto}(\alpha(S), x_m)$ with $\alpha'(S) < 0$ (higher pollution reduces the shape parameter, making the tail heavier) and fixed support $x_m > 0$. Moreover, suppose $\alpha > 1$ such that a finite first moment exists.

Proposition 2.2.3 (Pareto distribution decomposition).

Under Pareto shocks, the stochastic externality correction term can be rewritten as follows:

$$-\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] = \frac{\alpha'(S_2)}{\alpha(S_2)} \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) x_2 \ln \left(\frac{x_2}{x_m} \right) \right]$$

Proof. The Pareto PDF is $f(x; \alpha) = \alpha x_m^\alpha x^{-(\alpha+1)}$ and CDF is $F(x; \alpha) = 1 - \left(\frac{x_m}{x}\right)^\alpha$ for $x \geq x_m$. We directly apply the result in Proposition 2.2.1 on the CDF form of the stochastic externality term. $\square$

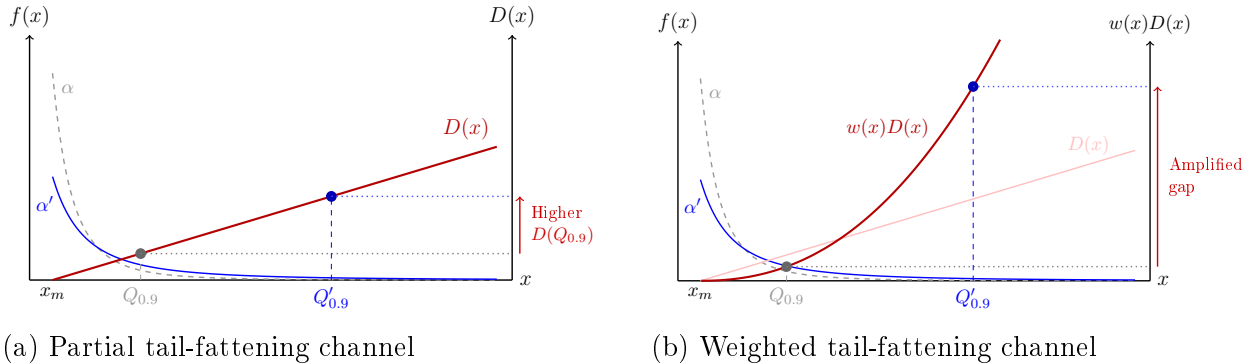


Figure 2: Pareto tail-fattening channel.

Endogenous Pareto shocks involve a single tail-fattening channel, illustrated in Figure 2. When the shape parameter α falls and the right tail grows heavier, each quantile of shocks shifts right, as shown in panel (a). But right-tail damages also carry a quantile-shift weight $w(x) \equiv x \ln(x/x_m)$, shown in panel (b), which grows without bound and places ever more weight on rare catastrophic events. Depending on functional forms, the expectation may diverge and the stochastic externality term become unbounded, echoing the "Dismal Theorem" of Weitzman (2009). The expectation is more "fragile" than the standard externality term. This paper does not focus on the Dismal Theorem, so we take only a brief detour. Consider $x \ln(x/x_m)$, ignoring the other terms in the expectation. If $\alpha > 1$, then $\mathbb{E}[x] = \frac{\alpha x_m}{\alpha - 1}$. For the expectation of $x \ln(x/x_m)$, note that $y \equiv \ln(x/x_m)$ follows an exponential distribution with rate α .

$$\mathbb{E} \left[x \ln \left(\frac{x}{x_m} \right) \right] = x_m \mathbb{E}[ye^y]$$

$\mathbb{E}[ye^y]$ is the derivative of the exponential distribution's moment generating function with respect to its argument s :

$$M_y = \mathbb{E}[e^{sy}] \implies \frac{\partial M_y}{\partial s} = \mathbb{E}[ye^{sy}]$$

Since $M_y = \alpha/(\alpha - s)$ as well,

$$M_y = \frac{\alpha}{\alpha - s} \implies \mathbb{E}[ye^y] = \frac{\partial M_y}{\partial s} \Big|_{s=1} = \frac{\alpha}{(\alpha - 1)^2}$$

Hence,

$$\mathbb{E} \left[x \ln \left(\frac{x}{x_m} \right) \right] = \frac{\alpha x_m}{(\alpha - 1)^2} = \frac{\mathbb{E}[x]}{\alpha - 1}$$

As $\alpha \rightarrow 1$ and the tails become heavier, $\mathbb{E}[x \ln(x/x_m)]$ diverges faster than $\mathbb{E}[x]$, so the stochastic externality term may become unbounded before the standard externality term. Even if the Dismal Theorem does not hold with standard externalities¹, it may still hold with stochastic externalities. We leave a fuller treatment of stochastic externalities and the Dismal Theorem to future work. The main takeaway is that the assumed distribution of shocks affects not only the channels through which stochastic externalities operate but also their size relative to standard externalities, which they can dominate. This matters because the distribution of climate events is thought to be fat- or heavy-tailed (generalized Gamma, generalized Pareto, and so on). Whether aggregating all climate shocks into a single stochastic damage function yields a fat- or heavy-tailed distribution is an open question for future research.

Log-normal shocks. Let $x \sim \text{LogNormal}(\mu(S), \sigma(S))$, meaning $\ln x \sim N(\mu(S), \sigma(S))$, with fixed support $x > 0$, $\mu'(S) > 0$ (higher pollution raises the location of log-shocks) and $\sigma'(S) > 0$ (higher pollution raises their dispersion). The log-normal is a natural bridge between the two previous examples: it is the exponential of a normal, so it inherits the two-parameter structure of the normal case, yet it is right-skewed and fat-tailed relative to the normal, like the Pareto. Because $\ln x$ is normal, the score takes the same form as in the normal case with x replaced by $\ln x$, but the change of variables introduces a level weight x in every channel.

Proposition 2.2.4 (Log-normal distribution decomposition).

Under log-normal shocks, the stochastic externality correction term decomposes into a geometric

¹See for example Nordhaus (2009) or Anthoff and Tol (2022), who present the Dismal Theorem as a real concern but show that its applicability depends on the assumptions and the modeling context.

mean-shift channel, a level-weighted risk-aversion channel, a level-weighted damage-curvature channel, and a Jensen (variance-raises-the-mean) channel:

$$\begin{aligned}
-\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= \mu'(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) x_2 \right] \\
&\quad - \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right)^2 x_2^2 \right] \\
&\quad + \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 D(\cdot)}{\partial x_2^2} Y(k_2) x_2^2 \right] \\
&\quad + \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) x_2 \right]
\end{aligned}$$

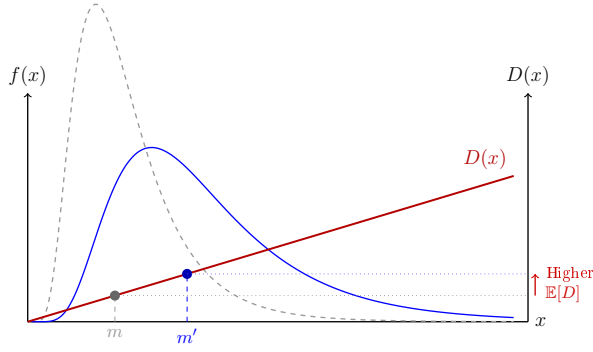
Proof. See Appendix C for full proof. $\square$

The first three channels mirror the normal decomposition of Proposition 2.2.2 but each carries an extra factor of the shock level x_2 (or x_2^2). This level weighting is exactly the Jacobian $dx_2/d(\ln x_2) = x_2$ of the log transformation and reflects that, unlike additive normal shocks, log-normal shocks act *multiplicatively*: a given proportional worsening of the climate translates into larger absolute damages in already-bad states. The fourth channel has no counterpart in the normal case. It arises because the mean of a log-normal, $\mathbb{E}[x_2] = e^{\mu(S_2) + \sigma(S_2)^2/2}$, increases with the dispersion $\sigma(S_2)$, so a pure increase in variability shifts the shock distribution rightward and raises expected damages even when the median is held fixed.

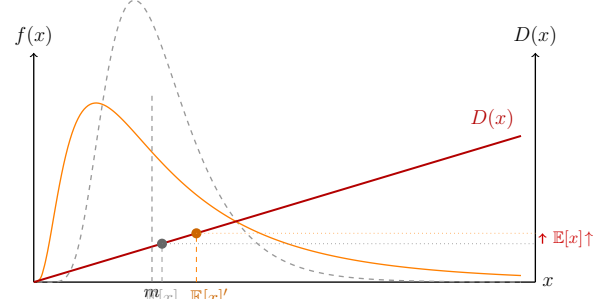
Figure 3 illustrates the mechanisms. The geometric mean-shift channel in panel (a) raises the tax when μ increases, since the entire multiplicative shock distribution scales upward. The Jensen channel in panel (b) is new relative to the normal case: a pure increase in dispersion σ raises the mean of x itself, so expected damages rise even with the median held fixed. The level-weighted risk-aversion and damage-curvature channels are shown jointly in panel (c): they operate as in the normal case but are amplified in the right tail by the quantile-shift weight $w(x) \equiv x \ln x$, which places growing weight on rare, severe realizations. Panel (d) previews the Dismal-Theorem question to which we now turn.

Endogenous log-normal shocks raise the same concern as the Pareto case: can the stochastic externality become unbounded? The dispersion channel weights damages by $x_2(\ln x_2 - \mu)/\sigma$, which grows like $x_2 \ln x_2$ in the right tail, the same order as the Pareto weight $x_2 \ln(x_2/x_m)$. The crucial difference is the density. Writing $y \equiv \ln x \sim N(\mu, \sigma^2)$ and using the normal moment generating function $M(s) = e^{\mu s + \sigma^2 s^2/2}$, every moment of the form $\mathbb{E}[x^k (\ln x)^j]$ is a Gaussian integral against e^{ky} and is therefore finite for all k , with $\mathbb{E}[x^k] = e^{k\mu + k^2 \sigma^2/2}$. Focusing on the raw dispersion weight as we did for the Pareto case, and using $\mathbb{E}[ye^y] = (\mu + \sigma^2)e^{\mu + \sigma^2/2}$,

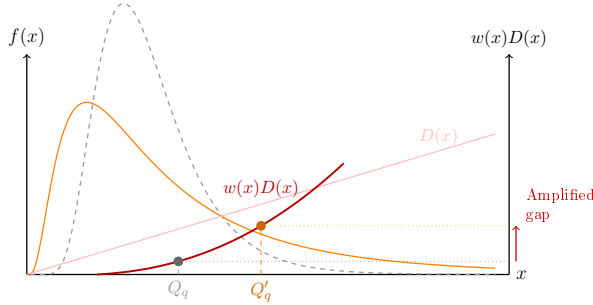
$$\mathbb{E} \left[x \frac{\ln x - \mu}{\sigma} \right] = \sigma \mathbb{E}[x] = \sigma e^{\mu + \sigma^2/2}.$$



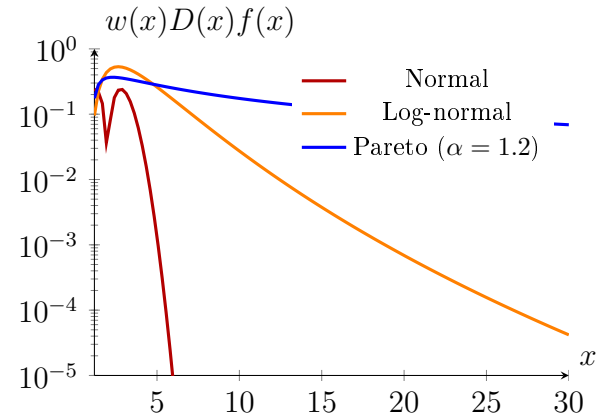
(a) Geometric mean-shift channel



(b) Jensen (variance) channel



(c) Level-weighted tail-fattening



(d) Tail of the correction's integrand

Figure 3: Stochastic externality channels for log-normally distributed shocks. Panels (a)–(c) illustrate the three economic mechanisms; panel (d) compares the right-tail contribution $w(x)D(x)f(x)$ of the stochastic externality across distributions on a log scale, showing the heavy but integrable log-normal tail sitting between the thin normal tail and the slowly-decaying Pareto tail.

Whereas the analogous Pareto quantity $\mathbb{E}[x \ln(x/x_m)] = \mathbb{E}[x]/(\alpha - 1)$ diverges as $\alpha \rightarrow 1$, the log-normal quantity $\sigma \mathbb{E}[x]$ is finite for every σ . Under polynomially growing damages, the log-normal stochastic externality therefore never literally diverges: there is no parameter value at which a moment ceases to exist, and the Dismal Theorem in the strict sense of Weitzman (2009) does not bite. This is consistent with the well-known observation that a log-normal climate sensitivity yields a finite expected marginal utility where a genuinely fat-tailed one need not.

Two caveats temper this reassuring conclusion. First, although finite, the correction grows *super-exponentially* in σ : for damages of order x^d the level-weighted variance channels scale like $e^{(d+1)^2\sigma^2/2}$ and $e^{(d+2)^2\sigma^2/2}$, dominating the standard Pigouvian term (of order $e^{(d-1)^2\sigma^2/2}$) as climate variability rises. In practice the stochastic externality can thus swamp the Pigouvian term well before any formal divergence. Second, the log-normal moment generating function is itself infinite, $\mathbb{E}[e^{sx}] = \infty$ for all $s > 0$, so the convergence result hinges on damages growing only polynomially; exponentially growing damages would make even the log-normal correction diverge. The log-normal therefore occupies an intermediate, “near-dismal” position between the thin-tailed symmetric normal, where stochastic externalities are mild, and the heavy-tailed Pareto, where they can genuinely diverge, underscoring once more that the assumed shock distribution governs not only the channels through which stochastic externalities operate but also their magnitude relative to the standard term.

In the next section, we generalize the three-period framework to an arbitrary number of periods. These general results do not depend on the specific way we model the externality. We chose the standard neoclassical model with multiplicative damages on output to build intuition, but we could just as well have modeled the externality as arising from polluting consumption instead of polluting output. The specific form would differ, but a stochastic externality would still arise, with the same channels at play.²

²Polluting consumption, for example, would affect the Euler condition through marginal utility rather than the marginal product of output. There would still be a standard externality term and a stochastic externality term, whose channels depend on the assumed distribution of shocks.

3 Multiple period model

Now consider an infinite-horizon problem in which a pollution stock accumulates, with the past stock depreciating linearly at rate ϕ . The sequential problem can be written in integral form as follows, with the pdf of shocks depending on the pollution stock. Let $x^t \equiv (x_0, x_1, \dots, x_t)$ be the history of shocks up to t and $\beta \in (0, 1)$ the discount factor.

$$\max_{\{c_t(x^t), k_{t+1}(x^t), S_{t+1}(x^t)\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t \int \cdots \int u(c_t(x^t)) f(x_0; \theta(S_0)) \cdots f(x_t; \theta(S_t)) dx_0 \cdots dx_t \right\} \quad (2)$$

s.t.

$$\begin{aligned} c_t(x^t) + k_{t+1}(x^t) &= (1 - D(S_t, x_t))Y(k_t(x^{t-1})) + (1 - \delta)k_t(x^{t-1}) & \forall t \\ S_{t+1}(x^t) &= (1 - \phi)S_t(x^{t-1}) + Y(k_t(x^{t-1})) & \forall t \\ x_{t+1} &\sim f(x_{t+1}; \theta(S_{t+1}(x^t))) & \forall t \end{aligned}$$

As written, this sequential problem is unwieldy. Rewriting it recursively makes it more akin to the two-period problem. Although the standard recursive methods used in economics assume exogenous shocks following a Markov process (see for example (Stokey et al., 1989)), recursivity still holds here. The shocks follow a *Markov decision process* (MDP), in which the transition probability depends on the current state and action. With minor adjustments, the main results of Stokey et al. (1989) still hold³. We make the following standard assumptions.

Assumption 3.0.1.

Let Γ be the correspondence for the constraints of this problem mapping states $s \in \mathbb{S}$ (i.e. capital and pollution stock) to actions $a \in \mathbb{A}$ (i.e. output). Suppose Γ has a measurable selection, meaning there is a measurable function $h : \mathbb{S} \rightarrow \mathbb{A}$ such that $h(s) \in \Gamma(s) \quad \forall s \in \mathbb{S}$.

Assumption 3.0.2.

Let utility be $u : \mathbb{S} \times \mathbb{A} \rightarrow \mathbb{R}$ and the stochastic kernel be $P : (\mathbb{S} \times \mathbb{A}) \times \mathcal{S} \rightarrow [0, 1]$, where $\mathcal{S}$ is the σ -algebra over the state space $\mathbb{S}$. Let a plan $\pi \in \Pi$ be a sequence of actions $(\pi_0, \pi_1, \dots)$ such that $\pi_0 \in \mathbb{A}$ and $\pi_t : \mathbb{S}^t \rightarrow \mathbb{A}$ are measurable for all t . For each $s_0 \in \mathbb{S}$ and each $\pi \in \Pi(s_0)$, $u(s_t, \pi_t(s^t))$ is $\mu^t(s_0, ds^t; \pi)$ -integrable and the following limit exists.

$$u(s_0, \pi_0) + \lim_{T \rightarrow \infty} \sum_{t=1}^T \beta^t \int u(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi)$$

³See Miao (2014) for a simple introduction to MDP dynamic programming, or Bertsekas (1995) and Hernández-Lerma and Lasserre (1996) for a more formal treatment.

Assumption 3.0.3.

Let there be a collection of non-negative measurable functions $L_t : \mathbb{S} \rightarrow \mathbb{R}_+ \forall t$ such that

$$u(s_0, \pi_0) \leq L_0(s_0) , \quad u(s_t, \pi_t(s^t)) \leq L_t(s_0) , \quad \sum_{t=0}^{\infty} \beta^t L_t(s_0) < \infty \quad \forall \pi \in \Pi(s_0), s_0 \in \mathbb{S}$$

Assumption 3.0.1 ensures a feasible policy exists for any state, Assumption 3.0.2 ensures that the discounted sum of utilities is well-defined, and Assumption 3.0.3 is a technical condition that allows use of the dominated convergence theorem in the following lemma.

Lemma 3.0.1 (Recursive formulation).

The sequential problem (2) can be rewritten as the following recursive problem.

$$\begin{aligned} V(k, S, x) = \max_{c, k', S'} & \left\{ u(c) + \beta \int V(k', S', x') f(x'; \theta(S')) dx' \right\} \\ & s.t. \\ c + k' &= (1 - D(S, x))Y(k) + (1 - \delta)k \\ S' &= (1 - \phi)S + Y(k) \\ x' &\sim f(x'; \theta(S')) \end{aligned} \tag{3}$$

Proof. We follow Miao (2014) here, though other textbooks would serve as well. It suffices to show that this setting is a standard MDP dynamic programming problem. The state is the triple (k, S, x) and the action is (k', S') . The constraints can be rewritten so that c is an implicit function of (k', S') . By Assumptions 3.0.1 and 3.0.2, define the following discounted expected utility for any given state s_0 and policy π .

$$J(s_0, \pi) = u(s_0, \pi_0) + \lim_{T \rightarrow \infty} \left\{ \sum_{t=1}^T \beta^t \int u(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi) \right\}$$

For any state, define the following value function.

$$V(s) = \sup_{\pi \in \Pi(s)} \{J(s, \pi)\}$$

In this notation, $s_0 = (k_0, S_0, x_0)$, $s_t = (k_t, S_t, x_t) \forall t$, and $s = (k, S, x)$. Here $\pi \in \Pi(s)$ means the policy π is feasible given the initial state s , with feasibility determined by the constraint set.

Suppose that $V : \mathbb{S} \rightarrow \mathbb{R}$ is measurable and, given any $s \in \mathbb{S}$ and any $\epsilon > 0$,

$$\exists \pi^s \in \Pi(s) : J(s, \pi) + \epsilon \geq V(s)$$

such that π_0^s and $\pi_t^s(s_1^t)$ are measurable for all t . By Theorem 7.1.1 of Miao (2014), Assumptions 3.0.1-3.0.3 together with the above assumption imply that V satisfies:

$$V(s) = \sup_{a \in \Gamma(s)} \left\{ u(s, a) + \beta \int V(s') P(s, a; ds') \right\}$$

In our notation,

$$V(k, S, x) = \max_{(k', S') \in \Gamma(k, S, x)} \left\{ u(k, S, x) + \beta \int V(k', S', x') f(x'; \theta(S')) dx' \right\}$$

Alternatively, noting that c is an implicit function of (k, S, x) ,

$$\begin{aligned} V(k, S, x) = \max_{c, k', S'} & \left\{ u(c) + \beta \int V(k', S', x') f(x'; \theta(S')) dx' \right\} \\ & \text{s.t.} \\ c + k' &= (1 - D(S, x))Y(k) + (1 - \delta)k \\ S' &= (1 - \phi)S + Y(k) \\ x' &\sim f(x'; \theta(S')) \end{aligned}$$

By Theorem 7.1.2 of Miao (2014), this equation is also equivalent to $V(s) = \sup_{\pi \in \Pi(s)} \{J(s, \pi)\}$, so the recursive problem is equivalent to the sequential problem. $\square$

Although the shocks appear i.i.d., their dependence on last period's emissions creates persistence in the process. Because consumption affects the shock distribution only through the pollution stock, the state variable S is sufficient to capture the effect of past emissions on future shocks. This gives the process a Markov structure, which lets the problem be written recursively. Using this result, we derive the multi-period Euler condition, generalizing the three-period results to an arbitrary horizon.

Proposition 3.0.1 (Carbon tax with endogenous stochastic externalities).

The planner can decentralize the solution to problem (3) with the following carbon tax:

$$\tau_t = \underbrace{\beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right]}_{\text{"Standard" externality}} - \underbrace{\beta \sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \text{Cov} \left(\frac{V_{t+j+2}}{u'(c_{t+1})}, \frac{\partial \ln(f_{t+j+2})}{\partial S_{t+j+2}} \right)}_{\text{"Stochastic" externality}}$$

This tax can alternatively be expressed as follows:

$$\begin{aligned} \tau_t = & \underbrace{\beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right]}_{\text{"Standard" externality}} \\ & + \underbrace{\beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{x,t+j+2} Y_{t+j+2} \frac{\partial Q(p(x_{t+j+2}); \theta(S_{t+j+2}))}{\partial S_{t+j+2}} \right]}_{\text{"Stochastic" externality}} \end{aligned}$$

Proof. See Appendix D for full proof. □

This formula generalizes the three-period case to an arbitrary horizon. Because the pollution stock depreciates at rate ϕ , the effect of current emissions on future shocks is discounted by a factor $(1 - \phi)^j$ after j periods. That is, only $(1 - \phi)^j$ of emissions at time t will remain and harm the environment at time $t + j$. The first term is the standard Pigouvian term, equal to the discounted expected marginal damages of pollution; the second term is the stochastic externality term, capturing the effect of pollution on the distribution of shocks.

4 DICE-SE model

DICE-SE takes the annual, deterministic DICE at the core of DSICE and changes one equation, the damage function, so that emissions shift the distribution of climate losses and not only their mean (Cai & Lontzek, 2019). The model then prices the stochastic externality of Proposition 3.0.1: the part of the optimal carbon tax that exists because emissions load the climate die. We solve it by dynamic programming and compute the tax decomposition exactly, with no simulation. This section states the two modifications and the solution. Section 5 calibrates the shock, and Section 6 reports the premium.

4.1 Two modifications to DICE

The backbone is the six-state annual DICE-CJL model, with capital K_t , three carbon reservoirs ($M_{AT,t}, M_{U,t}, M_{L,t}$), and two temperatures ($T_{AT,t}, T_{OC,t}$). The carbon cycle and climate block are unchanged and appear in Appendix E. Two changes turn it into DICE-SE.

The damage function gains a stochastic loss. Net output is

$$Y_t = \frac{(1 - \Lambda_t) \tilde{Y}_t}{1 + a_1 T_{AT,t} + a_2 T_{AT,t}^2 + x_t - m(T_{AT,t})},$$

where $\tilde{Y}_t = A_t L_t^{1-\gamma} K_t^\gamma$ is gross output, Λ_t the abatement-cost share, $(a_1, a_2) = (0, 0.00284)$ the DICE damage coefficients, and $x_t \geq 0$ the aggregate climate loss as a share of gross world product. The shock enters at unit scale, since x_t is already a loss share, and $m(\cdot)$ is the recentering defined below. Setting $x_t = m(T) = 0$ recovers deterministic DICE. The denominator keeps damages strictly below one for any x_t , so a heavy-tailed shock is compatible with bounded output loss, which is what the Dismal-Theorem discussion of Section 2 requires. This is the theoretical damage function $1 - D(S, x) = 1/(1 + a_1 T + a_2 T^2 + x - m)$.

The loss share is a new state whose distribution depends on the climate,

$$x_{t+1} \sim F(\cdot; \theta(T_{AT,t+1})),$$

where F is the spliced family calibrated in Section 5 and $\theta(\cdot)$ maps the climate state to its parameters. The DICE climate system is predetermined one period ahead, so $T_{AT,t+1}$ is known at t . Tomorrow's loss distribution is fixed today and responds to today's emissions through the carbon-temperature block. This is the convention $x' \sim f(\cdot; \theta(S'))$ of Sections 2 and 3, with the pollution stock S read as the atmospheric temperature anomaly. The baseline treats x_t as conditionally independent over time; the measured persistence of the loss share is 0.15, and Section 6 shows it does not move the premium.

Two configurations separate the level of damages from their risk. Config A, the baseline, subtracts the conditional mean, $m(T) = \mathbb{E}[x \mid T]$, fitted by the quadratic $0.00113 + 0.00058 T - 0.00004 T^2$. Expected damages then stay at the DICE baseline, and any departure from deterministic DICE is pure risk. Config B sets $m \equiv 0$, treats extreme losses as absent from the DICE damage function, and adds them on top. It is an upper bound. Mean preservation is what makes the premium of Section 6 a clean risk object rather than a relabeled mean effect.

4.2 Solving the model and the exact tax decomposition

We solve DICE-SE by finite-horizon dynamic programming, integrating the shock out of the state space. Because x_t is conditionally independent given $\theta(T)$, the post-integration value $\widetilde{W}_t$ lives on the six deterministic states, and the shock enters only as a one-dimensional quadrature inside the Bellman operator. The recursion runs backward over 600 annual periods to a terminal block. A certainty-equivalent method would not serve here for two reasons. Under Config A it replaces x by $\mathbb{E}[x \mid T]$, cancels $m(T)$, and sees deterministic DICE, so it cannot respond to risk at all. And the premium is an asset-pricing object, a covariance between continuation values and the score, which the simulated certainty-equivalent method of Cai and Judd (2023) does not produce even as its dynamic-programming parent (Cai & Lontzek, 2019) does.

The optimal tax splits exactly as in Proposition 3.0.1,

$$\tau_s = \tau_s^{\text{std}} + \tau_s^{\text{stoch}}.$$

The standard term τ_s^{std} is the discounted marginal damage of the physical channel, present already in deterministic DICE. The stochastic-externality term is the score-weighted covariance

$$\tau_s^{\text{stoch}} = -\beta \sum_{j \geq 0} \beta^j \Lambda_{s,j} \text{Cov}_s \left(\frac{V_{s+j+1}}{u'(c_s)}, \frac{\partial \ln f_{s+j+1}}{\partial S_{s+j+1}} \right), \quad (4)$$

where $\Lambda_{s,j}$ propagates a date- s emission pulse to the date- $(s+j)$ distribution-shifting state through the carbon-temperature system. The code evaluates (4) exactly. Rather than simulate the covariance, it accumulates the per-period score injections along a second backward recursion, a score-channel costate defined on the same states, so the whole decomposition carries no simulation noise. A within-model identity confirms the computation: at an interior abatement rate the total tax equals the marginal abatement cost divided by the damage denominator, and the baseline solution satisfies it to 9×10^{-8} at 2005.

5 Calibrating the stochastic damage function

The theory needs two objects: a family $F(x; \theta)$ for the aggregate loss and a map $\theta(S)$ from the climate state to its parameters, satisfying Assumption ???. Everything about the microstructure of climate risk reaches the carbon tax only through this pair. We estimate it on 1980–2024 disaster losses as a single spliced family whose arrival rate and body shift with warming and whose tail is heavy.

5.1 One distribution for many hazards

The aggregate loss is the sum of hazard-level losses, $x_t = \sum_h L_{h,t}$, over cyclones, floods, heat, drought, and wildfire. Each hazard is a compound frequency-severity process, the workhorse of catastrophe modeling and non-life insurance, and extreme value theory disciplines its severity: by the Pickands-Balkema-de Haan theorem, exceedances over a high threshold converge to a generalized Pareto distribution, so each severity tail is $\text{GPD}(\zeta_h, \xi_h)$ whatever

the hazard’s physical nature (Coles, 2001; Embrechts et al., 1997). Summing across hazards does not destroy this structure. For heavy tails the sum inherits the tail of its heaviest component, $\xi = \max_h \xi_h$ (Embrechts et al., 1997). A single family is therefore not a convenient approximation. It is the asymptotically correct description of aggregate risk.

We use a spliced form: a lognormal body below a fixed threshold $\bar{u}$ carrying weight $1 - \pi(S)$, and a generalized Pareto tail above it,

$$F(x; \theta(S)) = \begin{cases} (1 - \pi(S)) B(x; \theta_B(S)) & x \leq \bar{u}, \\ (1 - \pi(S)) + \pi(S) \text{GPD}(x - \bar{u}; \varsigma(S), \xi) & x > \bar{u}, \end{cases}$$

with parameter vector $\theta(S) = (\theta_B(S), \pi(S), \varsigma(S), \xi)$ and a fixed threshold that preserves the fixed-support requirement of Assumption ???. Each theoretical channel has a counterpart here. The body location matches the mean-shift channel of Proposition 2.2.2, the exceedance probability π and the tail scale ς match the dispersion channels, and the tail index ξ is the Pareto tail-fattening channel that Section 2 flags as able to dominate the tax.

5.2 Data and estimates

We estimate $\theta(T)$ on global event-level losses from EM-DAT (13,320 climate events, 1980–2024), expressed as shares of gross world product and adjusted for exposure growth (Pielke et al., 2008), with the NOAA billion-dollar archive as a comparison (NOAA National Centers for Environmental Information (NCEI), 2024). Three channels move with warming, and the data identify them unequally.

Frequency rises sharply. The arrival rate of large events grows by a factor $e^{0.59} \approx 1.8$ per degree of warming, with a 95% confidence interval of $[1.4, 2.3]$, estimated by Poisson regression of annual exceedance counts on temperature. This is the exceedance probability $\pi(T)$, and it climbs from about 0.18 at 0.8°C to 0.72 at 4°C.

The body shifts up. The median annual loss share roughly doubles per degree, with a body-location slope $\hat{\mu}_1 = 0.68$ ($p = 0.004$). This slope is sensitive to the sample window, ranging over $[0.2, 1.1]$ as the start year moves, a range we carry into the results as a sensitivity.

The tail is heavy, and in the aggregate loss data it does not visibly respond to warming. The fitted tail index is $\xi = 0.68$, with a bootstrap interval of $[0.54, 0.82]$, heavy enough that the loss share has a finite mean but an infinite variance. The response of the tail scale to warming is not identified from aggregate losses. Its reduced-form estimate is negative, $\hat{d} = -0.45$, which reflects composition and the historical decline in vulnerability rather than physics. A hazard-by-hazard physical program confirms the reading. Using basin sea-surface temperature for cyclones, antecedent precipitation for floods, and local temperature for heat, the per-event severity response is a small and imprecise positive once the time trend is absorbed, not the negative the pooled regression returns. We therefore set the tail-scale response to zero in the baseline, a conservative choice, and carry a physically disciplined positive variant, $d \in \{0.28, 0.53\}$, built from the same hazard program.

The fitted family passes a probability-integral-transform backtest, with a Kolmogorov-Smirnov p -value of 0.73, and matches external anchors: the fitted mean loss of 0.18% of gross world

product sits between attributable-damage and total-loss reinsurance estimates for the recent decade.

5.3 What the data can and cannot say

Three caveats bound the calibration. Adaptation comes first: $\theta(T)$ estimated from history is a reduced form along the observed adaptation path, so it prices hazard intensification net of the vulnerability decline that accompanied it, and a vulnerability-frozen calibration would be steeper. Attribution comes second: warming is nearly linear in time over 1980–2024, so the data cannot cleanly separate a temperature effect from a coincident trend, and the causal reading leans on the detection-and-attribution literature (Stott et al., 2016) rather than on the regression alone. Sample length comes third: the tail rests on few large observations, which is why Section 6 reports the tax as a function of ξ rather than at a point estimate. The tail index is the object the theory makes decisive and the data pin down least. Disciplining its warming response from physics, rather than from the short loss record, is the natural next step, and we return to it below.

6 The stochastic-externality premium

At the 2005 calibration the optimal carbon tax is \$36.50 per ton of carbon, of which the stochastic externality accounts for \$2.13, or 5.8%. The premium is modest at the baseline and grows steeply in the tail index, and under recursive preferences it roughly doubles. This section reports the decomposition, its comparative statics, and the verification behind every number. All figures are 2005 US dollars per ton of carbon.

6.1 Baseline decomposition

Figure 4 shows the decomposition along the optimal path. The 2005 premium of \$2.13 rises in levels to \$3.82 by 2100, while its share of the tax falls toward 2% as the Pigouvian term accelerates with warming.

Two features govern how to read the number. First, under mean preservation the premium is not a net addition to the tax. The total 2005 tax exceeds deterministic DICE by only \$0.05, because the recentering $m(T)$ moves the mean loss into the Pigouvian term and lowers it by almost exactly the mean-shift content of the score channel. The decomposition (4) makes this relabeling visible instead of hiding it. Second, the premium is mostly that mean-shift content. Scaling down the shock’s dispersion alone, holding its mean sensitivity fixed, leaves \$1.9 of the \$2.13 standing. About nine-tenths of the 2005 premium is mean-shift and one-tenth, roughly \$0.2, is pure risk. With power utility and conditionally independent shocks of the calibrated size, risk aversion alone moves the tax little. The economic weight of the premium is in its distribution-shifting content, and above all in its growth in the tail index.

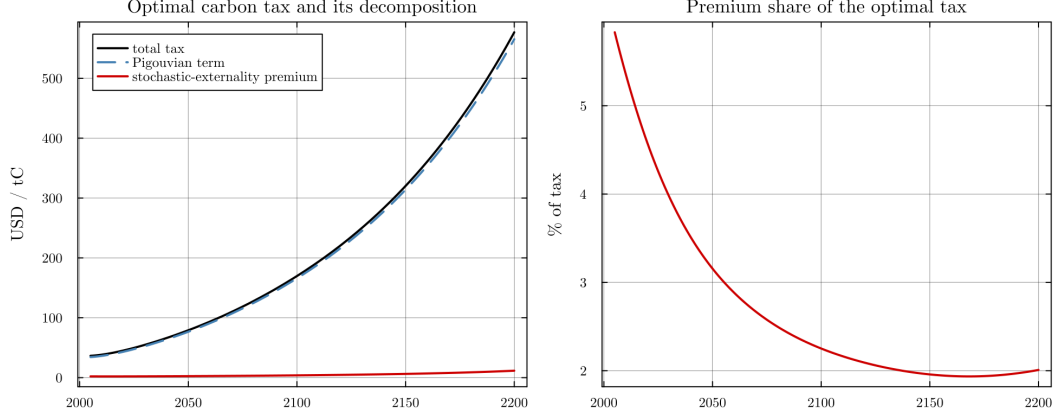


Figure 4: Baseline decomposition (Config A, $d = 0$). The optimal tax and its split into the Pigouvian term and the stochastic-externality premium along the optimal path.

6.2 The tail index is the comparative static that matters

The premium rises steeply and convexly in the tail index (Figure 5). Table 1 reports the full grid. Across the bootstrap range of ξ the 2005 premium runs from \$1.90 at $\xi = 0.54$ to \$2.51 at $\xi = 0.82$. At the US-pooled value $\xi = 1.05$, where the loss share has no finite mean, it reaches \$3.54, and the dynamic program still solves routinely where a certainty-equivalent method is undefined. The physically disciplined tail-scale variants raise the premium similarly, to \$2.91 at $d = 0.28$ and \$3.93 at $d = 0.53$. Config B, which adds the mean loss on top of DICE, raises the total tax by about \$2 but leaves the premium near \$2.13. The score channel prices the distribution's sensitivity to warming, not its level. The body-location window μ_1 mostly relabels between the two columns under mean preservation, moving the premium far less than ξ does.

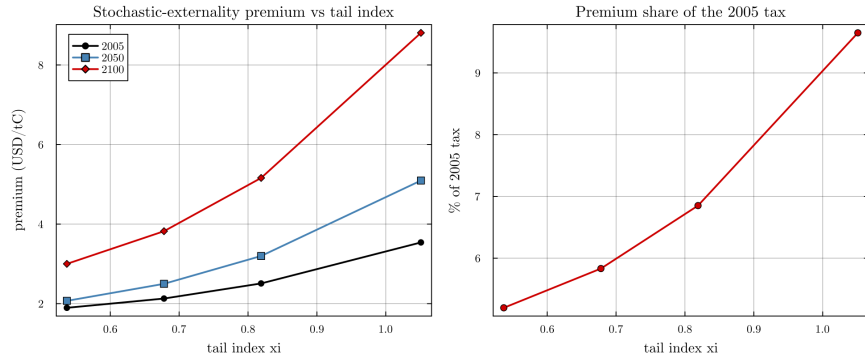


Figure 5: The stochastic-externality premium as a function of the tail index ξ , all other parameters at baseline.

Table 1: The 2005 decomposition and selected path values across the calibration grid. Config A unless stated; every variant recomputes its mean recentering. “Share” is the 2005 premium share of the tax in percent; the last column is atmospheric temperature in 2100.

Run	τ_{05}	pigou ₀₅	prem ₀₅	share	prem ₅₀	prem ₂₁₀₀	T_{2100}
Baseline: $d = 0$, $\xi = 0.678$	36.50	34.37	2.13	5.8	2.50	3.82	2.48
$d = 0.28$ (central physical variant)	36.58	33.67	2.91	8.0	4.71	9.84	2.48
$d = 0.53$ (high physical variant)	36.78	32.85	3.93	10.7	8.24	21.26	2.49
$\xi = 0.537$ (CI lower)	36.48	34.59	1.90	5.2	2.07	3.00	2.48
$\xi = 0.819$ (CI upper)	36.61	34.10	2.51	6.9	3.20	5.16	2.48
$\xi = 1.051$ (US pooled)	36.68	33.14	3.54	9.6	5.09	8.81	2.48
$\mu_1 = 0.21$ (window lower)	36.44	34.78	1.66	4.6	2.56	4.34	2.48
$\mu_1 = 0.86$ (window upper)	36.51	34.26	2.25	6.2	2.42	3.69	2.48
Config B: additive, $m = 0$	38.47	36.34	2.13	5.5	2.49	3.82	2.48
Persistence $\rho = 0.146$	36.34	34.24	2.10	5.8	2.45	3.74	2.48

6.3 Recursive preferences double the premium

Because the stochastic externality is a risk-pricing term, recursive utility is the natural benchmark (Cai & Lontzek, 2019; Epstein & Zin, 1989). Table 2 reports the premium on a grid over risk aversion γ and the elasticity of intertemporal substitution ψ , computed by solving the model twice, once with $\theta(T)$ active and once frozen at the 2005 climate, a route that does not depend on any recursive score formula. At the DSICE benchmark, $\gamma = 10$ and $\psi = 1.5$, the premium is \$4.18 against \$2.13 under power utility, and the 2005 tax is \$98.6. The amplification runs through the elasticity of intertemporal substitution, not through risk aversion. The premium is flat in γ at fixed ψ , and it doubles as ψ rises from 0.5 to 1.5. A higher elasticity lowers the effective consumption discount rate, which raises the whole tax, and the premium scales with it. Risk aversion moves the premium little, consistent with the small pure-risk content of the baseline.

Table 2: Epstein-Zin stochastic-externality premium over (γ, ψ) , Config B, computed by the model-twice route. Tax and premium in 2005 US dollars per ton of carbon; “share” is the 2005 premium share of the tax. The pair $(\gamma, \psi) = (2, 0.5)$ is the power-utility corner; $(10, 1.5)$ is the DSICE benchmark.

γ	ψ	τ_{05}	pigou ₀₅	prem ₀₅	share	prem ₂₁₀₀
2	0.5	39.88	37.76	2.12	5.31	3.64
2	1.5	98.62	94.44	4.18	4.24	9.77
10	0.5	39.89	37.76	2.12	5.32	3.67
10	1.5	98.63	94.44	4.18	4.24	9.79
10	2.0	116.57	111.74	4.82	4.14	11.99
15	1.5	98.63	94.44	4.19	4.25	9.80

6.4 Verification

Every number survives a five-part check. With risk switched off, the dynamic program reproduces the verified deterministic DICE solution to 2×10^{-6} in welfare and to 0.3% in the 2005 tax, so any residual gap is approximation error and not specification. Solving the model twice, active against frozen at the 2005 climate, and differencing the carbon prices reproduces the score-based premium to 1.3% at 2005. The frozen-climate premium is exactly zero, as it must be. Refining the quadrature and raising the polynomial degree move the premium by at most 0.4%. Adding the measured persistence of 0.15 changes it by 0.1%.

6.5 Discussion

The exercise gives a first empirically disciplined estimate of the stochastic-externality premium, and it locates the economic action precisely. At the calibrated shock and with power utility, the premium is 5.8% of the tax and is mostly a relabeling of the mean-damage channel. Its distinctive content, the pricing of a warming-driven change in the shape of the loss distribution, is small at the baseline because the aggregate loss data do not identify a warming response in the tail. The premium becomes large through the two channels the theory anticipates. It more than doubles as the tail index moves from its lower confidence bound to the US-pooled value, and it doubles again under recursive risk pricing at the DSICE benchmark.

The weight of the result therefore rests on a single object the data pin down least, the tail index and its response to warming. The honest statement is that the aggregate loss record cannot say whether warming fattens the tail. The physical route can. Because the aggregate tail is dominated by its heaviest hazard, a synthetic-cyclone route disciplines the tail response from that hazard and reads a warming-dependent tail index off a large simulated catalogue rather than off the short loss series. Whether the die is loaded through frequency alone, as the identified baseline has it, or through a fattening tail as well, is the empirical question on which the size of the premium turns, and it is answerable with hazard-model data where the loss record is silent.

7 Conclusion

In this paper, we develop a framework for analyzing the optimal correction of stochastic externalities, defined as the effect of pollution on the distribution of future climate shocks. This reflects the "distributional shift" perspective of climate risk, according to which higher CO₂ concentrations (or other climate variables) make extreme events not only more frequent but also more severe and more unpredictable. This perspective is well-documented in the Earth sciences and is beginning to be empirically assessed in economics, yet its implications for the social cost of carbon and optimal climate policy have not been formally characterized. This paper fills this gap and offers avenues for further research.

Working first in a tractable three-period neoclassical model, we derive a modified optimal carbon tax that decomposes into two terms: the standard Pigouvian term and a novel stochastic externality term. The latter can be expressed as the covariance between the ratio of utilities and the score of the damage shock distribution, or equivalently as the discounted expectation of marginal damage shocks weighted by the change in the shock distribution's quantile function induced by the pollution-driven distributional shift. The framework nests the standard Pigouvian tax as a special case when the damage distribution is exogenous. Under normal shocks with pollution-dependent mean and variance, the stochastic externality decomposes into three economically interpretable channels: a mean-shift channel, a risk-aversion channel, and a damage-curvature channel. Under Pareto shocks, the correction collapses to a single tail-fattening channel whose magnitude can grow without bound as the tail becomes heavier, hinting that Weitzman's Dismal Theorem may be reinforced in the case of stochastic externalities. Indeed, even if the standard Pigouvian term remains finite, the stochastic externality term may not. The qualitative and quantitative results are sensitive to assumptions on the distribution of shocks and its parameters, highlighting the importance of rigorously calibrating the stochastic component of the model. We then generalize these results to an infinite-horizon recursive model.

To illustrate the quantitative importance of the stochastic externality, we implement these results in DICE-SE, a modified version of the DSICE model developed in Cai and Lontzek (2019) which is solved using a modified form of the SCEQ algorithm developed in Cai and Judd (2023). Although the full DSICE model also features TFP shocks and tipping points, we strip away these additional elements and focus solely on the baseline recursive DICE model it is based on. We then add a stochastic damage component to the initially deterministic damage function whose parameters shift with the climate stock (temperature or CO₂ concentration). In the present version of the paper, the distributional forms and parameters are only illustrative. However, Section 5 lays out in detail the methodology by which further work will discipline them, drawing on the climate science literature and the burgeoning empirical economic literature on the distribution-shifting channel of climate risk. Nevertheless, when we compare the standard externalities case to the endogenous externalities case, the numerical exercise reveals that the theoretical results are consistent with the DICE-SE simulation. Once properly calibrated, the model will be able to give a more accurate estimate of the stochastic externality premium. Since some types of climate events are hypothesized to follow heavy or fat-tailed distributions, there is a potential for a large stochastic externality premium. Whether this is indeed the case depends on the

assumption on the distribution of shocks.

This framework carries direct implications for climate policy. Standard integrated assessment models, which either treat climate damages as deterministic or merely focus on the timing of shocks may systematically underestimate the social cost of carbon. While the tipping points literature focuses on the possibility of large irreversible jumps in damages in the future, our distribution shifting mechanism instead focuses on the change in everyday climate shocks.

This work requires many improvements. Currently, the DICE-SE simulation is merely illustrative, using a normal distribution with arbitrary parameters rather than a carefully-chosen realistic shock distribution (e.g. a generalized Pareto or generalized extreme value family), as supported by the existing empirical literature. Future work will need to implement the calibration methodology of Section 5 to provide a more accurate estimate of the stochastic externality premium.

There are also a number of important extensions for future work, such as extending the framework to heterogeneous agents to allow the study of the distributional consequences of taking into account stochastic externalities. Moreover, it could be very interesting to investigate the interaction between mitigation and adaptation in this framework, as adaptation may reduce the need to internalize the stochastic externality through mitigation. Likewise, internalizing stochastic externalities may reduce the need for adaptation, acting as global insurance. Following Schumacher (2019), one might even argue that mitigating stochastic externalities is superior to adapting to them, since the former is essentially a public good while the latter is a private good, only a few being able to benefit from adaptation. In other words, a stochastic externality framework combined with an insurance component may also help us evaluate the relative merits of *ex post* mitigation ("prevention is better than cure") and of *ex ante* adaptation in the context of climate change.

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A Proof of Proposition 2.2.1

First, we solve the household's problem, which has the following Lagrangian.

$$\begin{aligned}\mathcal{L} = & u(r_0 k_0 - k_1 + (1 - \delta)k_0) + \beta \int_{x_1} u(r_1(x_1)k_1 - k_2(x_1) + (1 - \delta)k_1) f(x_1; \theta_1) dx_1 \\ & + \beta^2 \int_{x_1} \int_{x_2} u(r_2(x_2)k_2(x_1) + (1 - \delta)k_2(x_1)) f(x_1; \theta_1) f(x_2; \theta_2) dx_1 dx_2\end{aligned}$$

We assume the household is atomistic and therefore does not take into account the effect of their actions on the environment. Hence, we treat the pdf as exogenous in this problem. This yields the following first-order conditions.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial k_1} : u'(c_0) &= \beta \int_{x_1} u'(c_1)(1 - \delta + r_1) f(x_1; \theta_1) dx_1 \\ \frac{\partial \mathcal{L}}{\partial k_2} : u'(c_1) &= \beta \int_{x_2} u'(c_2)(1 - \delta + r_2) f(x_2; \theta_2) dx_2\end{aligned}$$

By the firm's optimality condition, we have $r_t = (1 - D(S_t, x_t))Y'(k_t)$. Hence,

$$\begin{aligned}u'(c_0) &= \beta \mathbb{E}_{x_1} [u'(c_1)(1 - \delta + (1 - D(S_1, x_1))Y'(k_1))] \\ u'(c_1) &= \beta \mathbb{E}_{x_2} [u'(c_2)(1 - \delta + (1 - D(S_2, x_2))Y'(k_2))]\end{aligned}$$

This is the standard Euler condition. The planner's problem yields the following FOCs.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial k_1} : u'(c_0) &= \beta \int_{x_1} u'(c_1)(1 - \delta + (1 - D(S_1, x_1))Y'(k_1)) f(x_1; \theta) dx_1 + \mu_2 Y'(k_1) \\ \frac{\partial \mathcal{L}}{\partial k_2} : u'(c_1) &= \beta \int_{x_2} u'(c_2)(1 - \delta + (1 - D(S_2, x_2))Y'(k_2)) f(x_2; \theta) dx_2 \\ \frac{\partial \mathcal{L}}{\partial S_1} : & -\beta \int_{x_1} u'(c_1) D_S(S_1) Y(k_1) f(x_1; \theta) dx_1 \\ & + \beta \int_{x_1} u(c_1) \frac{\partial f(x_1; \theta)}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} dx_1 \\ & + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial f(x_1; \theta)}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} f(x_2; \theta(S_2)) dx_1 dx_2 + (1 - \phi) \mu_2 = \mu_1 \\ \frac{\partial \mathcal{L}}{\partial S_2} : & -\beta^2 \int_{x_1} \int_{x_2} u'(c_2) D_S(S_2) Y(k_2) f(x_1; \theta) f(x_2; \theta) dx_1 dx_2 \\ & + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial f(x_2; \theta)}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2} f(x_1; \theta(S_1)) dx_1 dx_2 = \mu_2\end{aligned}$$

So that the terms in the brackets of the first-order condition with respect to c_0 are expressed as an expectation, we would like to rewrite the equation so that the second term is multiplied by the original pdf. To do so, we can use the property that $\frac{d \ln(f(x))}{dx} = \frac{f'(x)}{f(x)}$.

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial k_1} : u'(c_0) &= \beta \int_{x_1} u'(c_1)(1 - \delta + (1 - D(S_1, x_1))Y'(k_1))f(x_1; \theta)dx_1 + \mu_2 Y'(k_1) \\
\frac{\partial \mathcal{L}}{\partial k_2} : u'(c_1) &= \beta \int_{x_2} u'(c_2)(1 - \delta + (1 - D(S_2, x_2))Y'(k_2))f(x_2; \theta)dx_2 \\
\frac{\partial \mathcal{L}}{\partial S_1} : & -\beta \int_{x_1} u'(c_1)D_S(S_1)Y(k_1)f(x_1; \theta)dx_1 \\
& + \beta \int_{x_1} u(c_1) \frac{\partial \ln(f(x_1; \theta))}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} f(x_1; \theta(S_1))dx_1 \\
& + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial \ln(f(x_1; \theta))}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} f(x_1; \theta(S_1))f(x_2; \theta(S_2))dx_1 dx_2 + (1 - \phi)\mu_2 = \mu_1 \\
\frac{\partial \mathcal{L}}{\partial S_2} : & -\beta^2 \int_{x_1} \int_{x_2} u'(c_2)D_S(S_2)Y(k_2)f(x_1; \theta)f(x_2; \theta)dx_1 dx_2 \\
& + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial \ln(f(x_2; \theta))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2} f(x_1; \theta(S_1))f(x_2; \theta(S_2))dx_1 dx_2 = \mu_2
\end{aligned}$$

The last FOC can be written more compactly as follows.

$$\begin{aligned}
\mu_2 &= -\beta^2 \int_{x_1} \int_{x_2} u'(c_2)D_S(S_2)Y(k_2)f(x_1; \theta)f(x_2; \theta)dx_1 dx_2 \\
& + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial \ln(f(x_2; \theta))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2} f(x_1; \theta)f(x_2; \theta)dx_1 dx_2
\end{aligned}$$

From here, we can get the following modified Euler condition between c_0 and c_1 .

$$u_{c,0} = \beta \mathbb{E}_{x_1} \left[u_{c,1} \left(1 - \delta + \left[1 - D_1 - \beta \mathbb{E}_{x_2} \left[\underbrace{\frac{u_{c,2}}{u_{c,1}} D_{S,2} Y_2}_{\text{"Standard" ext.}} - \underbrace{\frac{u(c_2)}{u_{c,1}} \frac{\partial \ln(f(\cdot))}{\partial S_2}}_{\text{"Stochastic" ext.}} \right] Y_{k,1} \right) \right] \quad (5)$$

If we compare this to the household's Euler condition, we see that there is a wedge on the right-hand side. In order to decentralize the planner's solution, we need to set a carbon tax

$$\tau = \beta \mathbb{E}_{x_1, x_2} \left[\underbrace{\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2)}_{\text{"Standard" externality}} - \underbrace{\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(x_2; \theta(S_2)))}{\partial S_2}}_{\text{"Stochastic" externality}} \right]$$

We can also write this in a covariance formulation⁴.

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} - \underbrace{\beta \text{Cov} \left(\frac{u(c_2)}{u'(c_1)}, \frac{\partial \ln(f(x_2; \theta(S_2)))}{\partial S_2} \right)}_{\text{"Stochastic" externality}}$$

We can also write the Euler condition in cdf terms rather than pdf terms by rewriting the first-order condition with respect to S_2 with integration by parts⁵.

$$\begin{aligned} \int_{\underline{x}}^{\bar{x}} u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial f(x_2; \theta(S_2))}{\partial \theta} dx_2 &= u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} \Big|_{\underline{x}}^{\bar{x}} \\ &\quad - \int_{\underline{x}}^{\bar{x}} u'(c_2) \frac{\partial c_2}{\partial x_2} \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} dx_2 \end{aligned}$$

By the Leibniz integral rule, the derivative of the cdf with respect to its parameter is zero at the boundaries. Therefore,

$$\int_{\underline{x}}^{\bar{x}} u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial f(x_2; \theta(S_2))}{\partial \theta} dx_2 = - \int_{\underline{x}}^{\bar{x}} u'(c_2) \frac{\partial c_2}{\partial x_2} \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} dx_2$$

From the resource constraint, $\frac{\partial c_2}{\partial x_2} = -D_x(S_2, x_2)Y(k_2)$.

$$\int_{\underline{x}}^{\bar{x}} u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial f(x_2; \theta(S_2))}{\partial \theta} dx_2 = \int_{\underline{x}}^{\bar{x}} u'(c_2) D_x(S_2, x) Y(k_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} dx_2$$

⁴ $\frac{\partial \ln(f(\cdot))}{\partial \theta}$ is the score and $\mathbb{E}[S_\theta] = 0$. Hence, the second term in $\text{Cov}(x, y) = \mathbb{E}[xy] - \mathbb{E}[x] \mathbb{E}[y]$ cancels out.

⁵ Here, $u = u(c_2) \frac{\partial \theta(S_2)}{\partial S_2}$ and $dv = \frac{\partial f(\cdot)}{\partial \theta} dx$. Moreover, note that $\frac{\partial f(\cdot)}{\partial \theta} = \frac{\partial^2 F(\cdot)}{\partial x \partial \theta}$.

If we multiply and divide by $f(x_2; \theta(S_2)) \equiv \frac{\partial F(x_2; \theta(S_2))}{\partial x_2}$, we can rewrite equation (5) as:

$$u_{c,0} = \beta \mathbb{E}_{x_1} \left[u_{c,1} \left(1 - \delta + \left[1 - D_1 - \beta \mathbb{E}_{x_2} \left[\underbrace{\frac{u_{c,2}}{u_{c,1}} D_{S,2} Y_2}_{\text{"Standard" ext.}} - \underbrace{\frac{u_{c,2}}{u_{c,1}} D_x(S_2, x_2) Y(k_2) \frac{\frac{\partial F(x_2; \theta(S_2))}{\partial S_2}}{\frac{\partial F(x_2; \theta(S_2))}{\partial x_2}}}_{\text{"Stochastic" externality}} \right] \right] Y_{k,1} \right) \right]$$

There are three parts to this new term: the familiar stochastic discount factor, the marginal damages of stochastic shocks on output, and the change of the cdf over the pdf. This last part is hard to interpret, so we might want to rewrite it. The quantile function is defined as:

$$Q(p; \theta(S_2)) = F^{-1}(p; \theta(S_2))$$

From here, we can also write a cdf as follows.

$$F(Q(p; \theta(S_2)); \theta(S_2)) = p$$

If we differentiate both sides by S_2 and rearrange terms,

$$\frac{\frac{\partial F(Q(p; \theta(S_2)); \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}}{f(Q(p; \theta(S_2)); \theta(S_2))} = - \frac{\partial Q(p; \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}$$

If we set $x = Q(p; \theta(S_2))$ in the cdf,

$$\frac{\frac{\partial F(x; \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}}{f(x; \theta(S_2))} = - \frac{\partial Q(p; \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}$$

Hence, a different way to write equation (5) is:

$$u_{c,0} = \beta \mathbb{E}_{x_1} \left[u_{c,1} \left(1 - \delta + \left[1 - D_1 - \beta \mathbb{E}_{x_2} \left[\underbrace{\frac{u_{c,2}}{u_{c,1}} D_{S,2} Y_2}_{\text{"Standard" ext.}} - \underbrace{\frac{u_{c,2}}{u_{c,1}} D_x(S_2, x_2) Y(k_2) \frac{\partial Q(p(x_2); \theta(S_2))}{\partial S_2}}}_{\text{"Stochastic" externality}} \right] \right] Y_{k,1} \right) \right]$$

In this case, the implied corrective tax for the CDF and quantile function formulation is:

$$\tau = \beta \mathbb{E}_{x_1, x_2} \left[\underbrace{\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2)}_{\text{"Standard" externality}} - \underbrace{\frac{u'(c_2)}{u'(c_1)} D_x(S_2, x_2) Y(k_2) \frac{\frac{\partial F(x_2; \theta(S_2))}{\partial S_2}}{\frac{\partial F(x_2; \theta(S_2))}{\partial x_2}}}_{\text{"Stochastic" externality}} \right]$$

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} + \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_x(S_2, x_2) Y(k_2) \frac{\partial Q(p(x_2); \theta(S_2))}{\partial S_2} \right]}_{\text{"Stochastic" externality}}$$

B Proof of Proposition 2.2.2

The normal PDF is $f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$. In that case,

$$\frac{\partial \ln(f(\cdot))}{\partial S_2} = \left[\frac{\mu'(S_2)}{\sigma(S_2)} z + (z^2 - 1) \frac{\sigma'(S_2)}{\sigma(S_2)} \right]$$

Here, $z \sim N(0, 1)$. Hence, we can rewrite the "risk endogeneity" term as follows.

$$\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] = \frac{\mu'(S_2)}{\sigma(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} z \right] + \frac{\sigma'(S_2)}{\sigma(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} (z^2 - 1) \right]$$

Because labor is assumed to be exogenous and the shock is at the last period,

$$c_2(x_2) = (1 - D(S_2, x_2))Y(k_2) + (1 - \delta)k_2 \implies \begin{cases} \frac{\partial c_2}{\partial x_2} = -\frac{\partial D(S_2, x_2)}{\partial x_2} Y(k_2) \leq 0 \\ \frac{\partial^2 c_2}{\partial x_2^2} = -\frac{\partial^2 D(S_2, x_2)}{\partial x_2^2} Y(k_2) \leq 0 \end{cases}$$

By Stein's lemma and assuming $u_c \geq 0$ as well as $u_{cc} \leq 0$,

$$\begin{aligned} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} z \right] &= \sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial c_2}{\partial x_2} \right] \leq 0 \\ \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} (z^2 - 1) \right] &= \sigma(S_2)^2 \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial c_2}{\partial x_2} \right)^2 + \frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 c_2}{\partial x_2^2} \right] \end{aligned}$$

Although $\frac{\partial^2 c_1}{\partial x^2}$ could be positive or negative depending on the convexity of damages, we are squaring it and therefore making the entire term positive. Since we assumed $u_{cc} \leq 0$, the first term in the second equation is negative. However, the second term's sign depends on $\frac{\partial^2 c_2}{\partial x_2^2} = -D_{xx}Y(k_2)$, the curvature of damages in x . If we rewrite the "risk endogeneity" term,

$$\begin{aligned} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= -\mu'(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right] \\ &\quad + \sigma'(S_2)\sigma(S_2) \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right)^2 \right] \\ &\quad - \sigma'(S_2)\sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 D(\cdot)}{\partial x_2^2} Y(k_2) \right] \end{aligned}$$

The Pareto PDF is $f(x; \alpha) = \alpha x_m^\alpha x^{-(\alpha+1)}$ for $x \geq x_m$. In that case,

$$\frac{\partial \ln(f(x; \alpha(S_2)))}{\partial S_2} = \frac{\alpha'(S_2)}{\alpha(S_2)} \left[1 - \alpha(S_2) \ln \left(\frac{x}{x_m} \right) \right]$$

Let us define $y \equiv \ln \left(\frac{x}{x_m} \right)$. It is known that $y \sim \exp(\alpha)$, so we have $y \sim \exp(\alpha(S_2))$. Hence,

$$\begin{aligned} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= \frac{\alpha'(S_2)}{\alpha(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} - \alpha(S_2) \frac{u(c_2)}{u'(c_1)} y \right] \\ &= \frac{\alpha'(S_2)}{\alpha(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \right] - \alpha'(S_2) \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} y \right] \end{aligned}$$

C Proof of Proposition 2.2.4

The log-normal PDF is $f(x; \mu, \sigma) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)$ for $x > 0$. Writing $y \equiv \ln x \sim N(\mu, \sigma^2)$ and $z \equiv \frac{\ln x - \mu}{\sigma} \sim N(0, 1)$, the score is identical in form to the normal case with x replaced by $\ln x$:

$$\frac{\partial \ln(f(\cdot))}{\partial S_2} = \frac{\mu'(S_2)}{\sigma(S_2)} z + (z^2 - 1) \frac{\sigma'(S_2)}{\sigma(S_2)}.$$

Rather than work directly with the score, it is cleaner here to start from the quantile form of the stochastic externality in Proposition 2.2.1. The log-normal quantile function is $Q(p; \theta(S_2)) = \exp(\mu(S_2) + \sigma(S_2)\Phi^{-1}(p))$, where Φ^{-1} is the standard normal quantile function. At a realization $x_2 = Q(p; \theta(S_2))$ we have $\Phi^{-1}(p) = \frac{\ln x_2 - \mu}{\sigma}$, so

$$\frac{\partial Q(p; \theta(S_2))}{\partial S_2} = Q(p; \theta(S_2)) [\mu'(S_2) + \sigma'(S_2)\Phi^{-1}(p)] = x_2 \left[\mu'(S_2) + \sigma'(S_2) \frac{\ln x_2 - \mu}{\sigma(S_2)} \right].$$

The extra factor x_2 is the Jacobian $dx_2/d(\ln x_2)$ of the log transformation. Substituting into the quantile form of the stochastic externality term gives

$$- \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] = \mu'(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) x_2 \right] + \frac{\sigma'(S_2)}{\sigma(S_2)} \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) x_2 (\ln x_2 - \mu) \right].$$

The first term is the geometric mean-shift channel. For the second, change variables to $y = \ln x_2 \sim N(\mu, \sigma^2)$ and define

$$H(y) \equiv \frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) x_2 \Big|_{x_2=e^y} = g(x_2) x_2, \quad g(x_2) \equiv \frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2).$$

By Stein's lemma, $\mathbb{E}[H(y)(y - \mu)] = \sigma^2 \mathbb{E}[H'(y)]$, so the second term equals $\sigma'(S_2)\sigma(S_2) \mathbb{E}[H'(y)]$. Using $\frac{\partial c_2}{\partial x_2} = -\frac{\partial D(\cdot)}{\partial x_2} Y(k_2)$ and $dx_2/dy = x_2$,

$$H'(y) = x_2^2 g'(x_2) + x_2 g(x_2), \quad g'(x_2) = -\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right)^2 + \frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 D(\cdot)}{\partial x_2^2} Y(k_2).$$

Collecting the resulting terms yields the four channels of Proposition 2.2.4:

$$\begin{aligned}
-\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= \mu'(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) x_2 \right] \\
&\quad - \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right)^2 x_2^2 \right] \\
&\quad + \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 D(\cdot)}{\partial x_2^2} Y(k_2) x_2^2 \right] \\
&\quad + \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) x_2 \right].
\end{aligned}$$

Finally, the closed form for the dispersion weight used in the Dismal-Theorem discussion follows from the normal moment generating function $M(s) = \mathbb{E}[e^{sy}] = e^{\mu s + \sigma^2 s^2/2}$. Since $\mathbb{E}[ye^y] = M'(1) = (\mu + \sigma^2)e^{\mu + \sigma^2/2}$,

$$\mathbb{E} \left[x \frac{\ln x - \mu}{\sigma} \right] = \frac{1}{\sigma} \mathbb{E}[e^y(y - \mu)] = \frac{1}{\sigma} \left[(\mu + \sigma^2)e^{\mu + \sigma^2/2} - \mu e^{\mu + \sigma^2/2} \right] = \sigma e^{\mu + \sigma^2/2} = \sigma \mathbb{E}[x].$$

D Proof of Proposition 3.0.1

The problem (3) has the following Lagrangian.

$$\begin{aligned}\mathcal{L} = & u(c) + \beta \int V(k', S', x') f(x'; \theta(S')) dx' - \lambda [c + k' - (1 - D(S, x))Y(k) - (1 - \delta)k] \\ & - \mu [S' - (1 - \phi)S - Y(k)]\end{aligned}$$

This yields the following first-order conditions.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c} : & u'(c) = \lambda \\ \frac{\partial \mathcal{L}}{\partial k'} : & \beta \int \frac{\partial V}{\partial k'} f(x'; \theta(S')) dx' = \lambda \\ \frac{\partial \mathcal{L}}{\partial S'} : & \beta \int \frac{\partial V}{\partial S'} f(x'; \theta(S')) dx' + \beta \int V(k', S', x') \frac{\partial f(x'; \theta(S'))}{\partial \theta} \frac{\partial \theta(S')}{\partial S'} dx' = \mu\end{aligned}$$

By the envelope condition,

$$\frac{\partial V}{\partial S'} = -\lambda' D_S(S', x') Y(k') + (1 - \phi)\mu'$$

If we substitute λ' ,

$$\frac{\partial V}{\partial S'} = -u'(c') D_S(S', x') Y(k') + (1 - \phi)\mu'$$

Substituting into the FOC with respect to S' ,

$$\begin{aligned}\mu = & -\beta \int [u'(c') D_S(S', x') Y(k') - (1 - \phi)\mu'] f(x'; \theta(S')) dx' \\ & + \beta \int V(k', S', x') \frac{\partial f(x'; \theta(S'))}{\partial \theta} \frac{\partial \theta(S')}{\partial S'} dx'\end{aligned}$$

Using $\frac{d \ln(f(x))}{dx} = \frac{f'(x)}{f(x)}$, we can put everything under the same expectation.

$$\mu = -\beta \int \left[u'(c') D_S(S', x') Y(k') - V(k', S', x') \frac{\partial \ln(f(x'; \theta(S')))}{\partial \theta} \frac{\partial \theta(S')}{\partial S'} - (1 - \phi) \mu' \right] f(x'; \theta(S')) dx'$$

By the envelope condition,

$$\begin{aligned} \frac{\partial V}{\partial k'} &= \lambda' (1 - \delta + (1 - D(S', x')) Y'(k')) + \mu' Y'(k') \\ &= \lambda' \left(1 - \delta + \left(1 - D(S', x') + \frac{\mu'}{\lambda'} \right) Y'(k') \right) \end{aligned}$$

Hence, after combining the first two FOCs,

$$\begin{aligned} u'(c) &= \beta \int u'(c') \left(1 - \delta + \left(1 - D(S', x') - \beta \int \left[\frac{u'(c'')}{u'(c')} D_S(S'', x'') Y(k'') \right. \right. \right. \\ &\quad \left. \left. \left. - \frac{V(k'', S'', x'')}{u'(c')} \frac{\partial \ln(f(x''; \theta(S''))}{\partial S''} \right. \right. \right. \\ &\quad \left. \left. \left. - (1 - \phi) \frac{\mu''}{u'(c')} \right] f(x''; \theta(S'')) dx'' \right) Y'(k') \right) f(x'; \theta(S')) dx' \end{aligned}$$

By induction, we can put every $u'(\cdot)$ term and V term together in a weighted infinite sum. Meanwhile, the $(1 - \phi)\mu$ term vanishes by the transversality condition for the condition stock since $\beta(1 - \phi) < 1$ and μ is bounded.

$$\begin{aligned} u'(c_t) &= \beta \int u'(c_{t+1}) \left(1 - \delta + \left(1 - D_{t+1} - \beta \int \left[\sum_{j=0}^{\infty} ((1 - \phi)\beta)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_S(S_{t+j+2}, x_{t+j+2}) Y(k_{t+j+2}) \right. \right. \right. \\ &\quad \left. \left. \left. - ((1 - \phi)\beta)^j \frac{V_{t+j+2}}{u'(c_{t+1})} \frac{\partial \ln(f(x_{t+j+2}; \theta(S_{t+j+2})))}{\partial S_{t+j+2}} \right] f(\cdot) \right) Y'(k_{t+1}) \right) f(x_{t+1}; \theta(S_{t+1})) dx_{t+1} \end{aligned}$$

Hence, the corrective tax for period t is:

$$\tau_t = \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \left[\frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} - \frac{V_{t+j+2}}{u'(c_{t+1})} \frac{\partial \ln(f_{t+j+2})}{\partial S_{t+j+2}} \right] \right]$$

In covariance form,

$$\tau_t = \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right] - \sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \text{Cov} \left(\frac{V_{t+j+2}}{u'(c_{t+1})}, \frac{\partial \ln(f_{t+j+2})}{\partial S_{t+j+2}} \right)$$

We can also write the stochastic term in CDF or quantile form. Consider the stochastic part of the tax for a given j :

$$\int V(k_{t+j+2}, S_{t+j+2}, x_{t+j+2}) \frac{\partial \theta(S_{t+j+2})}{\partial S_{t+j+2}} \frac{\partial f(x_{t+j+2}; \theta(S_{t+j+2}))}{\partial \theta} dx_{t+j+2}$$

Integration by parts with $u = V_{t+j+2} \frac{\partial \theta}{\partial S}$ and $dv = \frac{\partial f}{\partial \theta} dx$ yields:

$$\begin{aligned} &= V_{t+j+2} \frac{\partial \theta}{\partial S} \frac{\partial F(x_{t+j+2}; \theta)}{\partial \theta} \Big|_{\underline{x}}^{\bar{x}} \\ &\quad - \int_{\underline{x}}^{\bar{x}} \frac{\partial V_{t+j+2}}{\partial x_{t+j+2}} \frac{\partial \theta}{\partial S} \frac{\partial F(x_{t+j+2}; \theta)}{\partial \theta} dx_{t+j+2} \end{aligned}$$

Here, the first term vanishes by the boundary condition. By the envelope theorem,

$$\frac{\partial V(k, S, x)}{\partial x} = -u'(c) D_x(S, x) Y(k)$$

Therefore,

$$\int V_{t+j+2} \frac{\partial \theta}{\partial S} \frac{\partial f}{\partial \theta} dx_{t+j+2} = \int u'(c_{t+j+2}) D_x(S_{t+j+2}, x_{t+j+2}) Y(k_{t+j+2}) \frac{\partial \theta}{\partial S} \frac{\partial F(x_{t+j+2}; \theta)}{\partial \theta} dx_{t+j+2}$$

Multiplying and dividing by f , the CDF form of the tax is:

$$\tau_t = \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \left[\frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} - \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{x,t+j+2} Y_{t+j+2} \frac{\frac{\partial F(x_{t+j+2}; \theta(S_{t+j+2}))}{\partial S_{t+j+2}}}{\frac{\partial F(x_{t+j+2}; \theta(S_{t+j+2}))}{\partial x_{t+j+2}}} \right] \right]$$

Applying the quantile identity,

$$\frac{\frac{\partial F(x; \theta(S))}{\partial \theta} \frac{\partial \theta(S)}{\partial S}}{f(x; \theta(S))} = - \frac{\partial Q(p; \theta(S))}{\partial \theta} \frac{\partial \theta(S)}{\partial S}$$

Hence, the quantile form of the tax is:

$$\begin{aligned} \tau_t = & \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right] \\ & + \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{x,t+j+2} Y_{t+j+2} \frac{\partial Q(p(x_{t+j+2}); \theta(S_{t+j+2}))}{\partial S_{t+j+2}} \right] \end{aligned}$$

E DICE climate module

The carbon cycle and climate dynamics follow the standard DICE specification unchanged. The carbon cycle is governed by the linear three-reservoir system:

$$\begin{aligned} M_{\text{AT},t+1} &= b_{11} M_{\text{AT},t} + b_{21} M_{\text{U},t} + E_t + E_t^{\text{land}} \\ M_{\text{U},t+1} &= b_{12} M_{\text{AT},t} + b_{22} M_{\text{U},t} + b_{32} M_{\text{L},t} \\ M_{\text{L},t+1} &= b_{23} M_{\text{U},t} + b_{33} M_{\text{L},t} \end{aligned}$$

where $E_t = \sigma_t(1 - \mu_t)\tilde{Y}_t$ denotes industrial emissions (with σ_t the carbon intensity of output and $\mu_t \in [0, 1]$ the mitigation rate), and E_t^{land} captures exogenous land-use emissions. The climate responds to CO_2 concentrations through the temperature equations:

$$\begin{aligned} T_{\text{AT},t+1} &= T_{\text{AT},t} + c_1 \left[\frac{F_{2 \times \text{CO}_2}}{\ln 2} \ln \left(\frac{M_{\text{AT},t}}{M_{\text{AT}}^{\text{pre}}} \right) + F_t^{\text{EX}} - \lambda T_{\text{AT},t} - c_3 (T_{\text{AT},t} - T_{\text{OC},t}) \right] \\ T_{\text{OC},t+1} &= T_{\text{OC},t} + c_4 (T_{\text{AT},t} - T_{\text{OC},t}) \end{aligned}$$

where $F_{2 \times \text{CO}_2}$ is the forcing from doubling CO_2 , $M_{\text{AT}}^{\text{pre}}$ is the preindustrial atmospheric concentration, F_t^{EX} is exogenous non- CO_2 forcing, λ is the climate sensitivity parameter, and c_1, c_3, c_4 are heat transfer coefficients.

Tauchen discretization

We discretize the AR(1) shock process using $K = 20$ equally spaced grid points on $[-m\sigma_x, m\sigma_x]$ where $m = 3$, giving step size $\Delta = 2m\sigma_x/(K-1)$. The transition probabilities of the resulting Markov chain are:

$$P_{ij} = \Phi \left(\frac{x_j + \Delta/2 - \rho x_i}{\sigma_\varepsilon} \right) - \Phi \left(\frac{x_j - \Delta/2 - \rho x_i}{\sigma_\varepsilon} \right)$$

for interior states $j \in \{2, \dots, K-1\}$, where Φ denotes the standard normal CDF. The boundary states accumulate the tail probabilities: $P_{i1} = \Phi((x_1 + \Delta/2 - \rho x_i)/\sigma_\varepsilon)$ and $P_{iK} = 1 - \Phi((x_K - \Delta/2 - \rho x_i)/\sigma_\varepsilon)$. In the endogenous model, σ_ε is replaced by $\sigma(S_{\text{eff}})$ and ρx_i by $\rho x_i + \mu(S_{\text{eff}})$, yielding the state-dependent matrix $P_{ij}(S)$ of equation (6).

E.1 State-dependent Tauchen discretization

The Tauchen grid is constructed once, wide enough to accommodate the worst-case distribution at the upper bound $S_{\text{eff}}^{\max} = 5.0^\circ\text{C}$. The grid points $\{x_1, \dots, x_K\}$ remain fixed, but the transition matrix is recomputed at each period based on the realized effective pollution state:

$$P_{ij}(S) = \Phi\left(\frac{x_j + \Delta/2 - \rho x_i - \mu(S)}{\sigma(S)}\right) - \Phi\left(\frac{x_j - \Delta/2 - \rho x_i - \mu(S)}{\sigma(S)}\right) \quad (6)$$

This is the discrete analogue of the theoretical conditional density $f(x'; \theta(S'))$ where the parameter vector $\theta = (\mu, \sigma)$ varies with the pollution stock. Unlike the exogenous case where P is constant over time, $P(S_{\text{eff},t})$ evolves with the climate trajectory, making the Markov chain *non-stationary*.